

Seungyeon Yoo

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EDUCATION

SEOUL NATIONAL UNIVERSITY

Seoul, Korea | Sep. 2021 – Present

Ph.D. Candidate in Aerospace Engineering (GPA 4.04/4.3)

- Advisor: H. Jin Kim
- Research Interests: Robot AI, Vision-Based Navigation, Visual Representation Learning
- Scholarship: BK Excellent Research Talent Fellowship

KOREA UNIVERSITY

Seoul, Korea | Mar. 2014 – Feb. 2020

B.S. in Mechanical Engineering (GPA 4.33/4.5)

- Scholarship: National Scholarship for Science and Engineering (merit-based; awarded for all 8 semesters)
- Awards: Grand Prize, Capstone Design Competition (₩2.0M, 2017); Academic Excellence Award (2017)
- Certificates: Future Entrepreneurship Certificate; Next Generation Leader Training Certificate

PROFESSIONAL EXPERIENCES

SAMSUNG ADVANCED INSTITUTE OF TECHNOLOGY

Research Intern (Computer Vision Technology Unit at the AI Research Center)

Suwon, Korea | Jul. 2023 – Sep. 2023

SAMSUNG ELECTRONICS

Engineer (Manufacturing Core Technology Team at the Global Technology Center)

Suwon, Korea | Mar. 2020 – Jul. 2021

ILSA (DRIVER ASSISTANCE SYSTEM STARTUP TEAM)

Team Leader

Seoul, Korea | Jan. 2018 – Jun. 2019

EXTRACURRICULAR ACTIVITIES

INTERNATIONAL SPACE EDUCATION BOARD (ISEB)

Korea Representative Student

Baku, Azerbaijan | Oct. 2023

KOREA UNIVERSITY INTERNATIONAL STUDENT ASSISTANT (KUISA)

Personnel Leader / Group Leader

Seoul, Korea | Feb. 2017 – Dec. 2017

PROJECTS

AUTONOMOUS AND SWARM SYSTEMS USING VISION-AIDED NAVIGATION IN GNSS-DENIED AND JAMMING ENVIRONMENTS

Funded by Defense Acquisition Program Administration

Jan. 2025 – Present

LANGUAGE MODEL-DRIVEN INTERACTION AND SPATIAL LEARNING FOR PERSISTENT AUTONOMOUS ROBOT NAVIGATION

Funded by Samsung Science and Technology Foundation

Jan. 2025 – Present

HUMANOID ROBOT DEVELOPMENT FOR HUMAN-LIKE PERCEPTION, COMMUNICATION, AND LEARNING

Funded by Korea Ministry of Trade, Industry and Energy

Jan. 2025 – Dec. 2025

LEARNING-BASED ROAD SURFACE CONDITION ESTIMATION FOR VEHICLE SLIP SUPPRESSION

Funded by Hyundai Motor Company

Jan. 2023 – Dec. 2024

AUTONOMOUS DRIVING-ASSISTIVE ROBOT FOR FOLDABLE WHEELCHAIRS

Funded by Korea Ministry of Science and ICT

Aug. 2021 – Dec. 2022

ADDITIONAL ACADEMIC BACKGROUND

KANGWON SCIENCE HIGH SCHOOL (Government-designated selective program for gifted students in science and engineering)

OTHER INFORMATION

Military Service: Sergeant, Republic of Korea Army, Mar. 2015 – Dec. 2016